

Piyush Raj

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CAREER OBJECTIVE

Final-year Bachelor of Technology in Agricultural Engineering student with internship experience at **ICAR-IISWC** and **CFMTTI**. Skilled in **CATIA V5**, **3DEXPERIENCE**, **ANSYS**, and **Python**, with knowledge of tractor systems, farm machinery, and agricultural mechanization. Seeking a **Graduate Engineer Trainee** role to apply technical skills in agricultural equipment design and development.

TECHNICAL SKILLS

CAD & Design: CATIA V5, SolidWorks, 3DEXPERIENCE, 3D Modeling, Assembly Design
Engineering: Engineering Drawing, GD&T, ANSYS, Machine Design, Tractor Systems, Farm Machinery
Programming: Python, Arduino, Smart Agriculture Automation
Core Competencies: Agricultural Equipment Design, Soil & Water Conservation, Irrigation Engineering, Tractor Systems Analysis
Soft Skills: Leadership, Teamwork, Problem Solving, Technical Communication, Documentation & Reporting
Languages: Hindi (Native), English (Proficient)

INTERNSHIP EXPERIENCE

ICAR - Indian Institute of Soil and Water Conservation (IISWC)

Internship

Jun 2025 - Jul 2025

- Gained practical exposure to soil and water conservation, watershed management, and irrigation engineering practices through field activities.
- Conducted field surveys, technical observations, and data collection while documenting conservation practices through technical reports.

Central Farm Machinery Training & Testing Institute (CFMTTI)

Internship

Jun 2026 - Jun 2026

- Analyzed tractor engine, transmission, hydraulic, PTO, steering, and braking systems with practical exposure to their operation and maintenance.
- Participated in tractor testing, inspection, preventive maintenance, and performance evaluation of agricultural machinery.

PROJECTS

Tractor Components Design using CATIA V5

CATIA V5, 3D Design, Assembly, GD&T

Tools & Technologies: CATIA V5, 3D Part Design, Assembly Design, Engineering Drawing, GD&T. **Outcome:** Developed accurate 3D CAD models and assemblies with detailed engineering drawings, strengthening practical skills in tractor component design, assembly integration, and engineering documentation.

Smart Greenhouse Automation

Arduino, Environmental Sensors, Automation

Automated environmental monitoring and control system reducing manual intervention and improving greenhouse system responsiveness through Arduino-based sensor integration and automated control mechanisms.

Battery-Powered Lawn Mower

DC Motor, Battery Systems, Mechanical Design

Developed a compact battery-powered mowing system focused on energy efficiency, reliable operation, and practical grass-cutting performance with optimized mechanical design.

EDUCATION

Centurion University of Technology and Management, Odisha

2023 – Present | GPA: 8.61/10

B.Tech - Agricultural Engineering

Bihar School Examination Board (BSEB)

2021 – 2023 | 66.6%

Intermediate - Science

Central Board of Secondary Education (CBSE)

2020 – 2021 | 70%

Matriculation

CERTIFICATIONS & ACHIEVEMENTS

Certifications: NPTEL Certification in Micro Irrigation Engineering, Drone Workshop Certificate, Basic Python Programming Certificate, CATIA Certification

Achievements: CATIA V5 Tractor Component Design, Academic Excellence (CGPA 8.61), Tractor Testing & Maintenance Experience, Student Leadership